

Solution hints of Numerical problems based on Binomial, Poisson + Normal probs dist<sup>ns</sup>.

1.  $n = 3, \quad p = P(1 \text{ or } 6) = \frac{2}{6} = \frac{1}{3} \quad X: \text{No. of Success}$

$$q = \frac{2}{3}$$

$$X \sim B(n, p)$$

$$\text{Mean} = np = 3 \times \frac{1}{3} = 1$$

$$\text{Variance} = npq = 3 \times \frac{1}{3} \times \frac{2}{3} = \frac{2}{3}$$

2.  $n = 5, \quad q = \text{Prob (failing)} = \frac{1}{8}, \quad p = \frac{7}{8}$

$$p(x) = \binom{n}{x} p^x q^{n-x}; \quad x = 0, 1, \dots, n.$$

$$\begin{aligned} \text{i) } P(X \geq 4) &= p(4) + p(5) = \binom{5}{4} \left(\frac{7}{8}\right)^4 \left(\frac{1}{8}\right) + \binom{5}{5} \left(\frac{7}{8}\right)^5 \left(\frac{1}{8}\right)^0 \\ &= ( \quad ) \text{ simplify} \end{aligned}$$

$$\text{ii) } P(X=5) = p(5) = \binom{5}{5} \left(\frac{7}{8}\right)^5 \left(\frac{1}{8}\right)^0 = ( \quad ) \text{ simplify}$$

3.  $p = 0.15, \quad n = 7, \quad q = 0.85, \quad p(x) = \binom{n}{x} p^x q^{n-x}; \quad x = 0, 1, \dots, n.$

$$P(X \geq 3) = 1 - P(X \leq 2) = 1 - (p(0) + p(1) + p(2)) \quad (\text{simplify})$$

$$\text{Mean} = np = 7 \times 0.15, \quad \text{Variance} = npq = 7 \times 0.15 \times 0.85$$

4.  $p = 0.01\% = 0.0001, \quad n = 1000,$

$p(\text{small}), \quad n(\text{large}), \quad \text{apply Poisson approximation}$

$$\lambda = np = 1000 \times 0.0001 = 0.1$$

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, \dots, \infty$$

$$P(X \leq 2) = p(0) + p(1) + p(2) \quad (\text{simplify})$$

5.  $p = 0.001 (\text{small}), \quad n = 2000 (\text{large}), \quad \text{apply Poisson approx.}$

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, \dots, \infty, \quad \lambda = np = 2000 \times 0.001 = 2$$

$$(a) \quad P(X=3) = p(3) = \frac{e^{-2} (2)^3}{3!} \quad (\text{simplify}), \quad \text{Similarly}$$

$$(b) \quad P(X \geq 3) = 1 - P(X \leq 2) = 1 - (p(0) + p(1) + p(2))$$

$$(c) \quad P(X \leq 2) = p(0) + p(1) + p(2) \quad \text{simplify}$$

6.  $n = 300$ ,  $p = 0.005$ ,  $N = 1000$   
 (large) (small), apply Poisson approximation  
 $\lambda = np = 300 \times 0.005 = 1.5$

$$f(x) = Np(x) ; p(x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, \dots$$

i)  $f(0) = Np(0)$  (solve) (answer to the nearest integer)

$$ii) f(x > 2) = Np(x > 2) = N[1 - (p(0) + p(1))]$$

$$iii) f(x \leq 2) = Np(x \leq 2) = N[p(0) + p(1) + p(2)]$$

7.  $n = 250$ ,  $p = 0.01$ ,  $N = 100$   
 (large) (small)

$$\lambda = np = 250 \times 0.01 = 2.5, \text{ use Poisson approximation}$$

Sol<sup>n</sup>: Exactly similar to Q.No. 6.

8.  $X \sim N(0.7515, (0.0020)^2)$ , Given  $\mu = 0.7515$   
 $N = 1000$   $\sigma = 0.0020$

$$P(\text{brass plug accepted}) = P(0.752 - 0.004 \leq X \leq 0.752 + 0.004) \\ = P(0.748 \leq X \leq 0.756)$$

$$P(\text{brass plug rejected}) = 1 - P(0.748 \leq X \leq 0.756)$$

No. of brass plug rejected  $f(x) = Np(x)$

$$f(x) = 1000 \times p(\text{rejection}) = 1000 \times [1 - P(0.748 \leq X \leq 0.756)]$$

9.  $X \sim N(\mu, \sigma^2)$ ;  $\mu = 100$ ,  $\sigma = 2$  Solve it.

$$P(98 < X < 102) = ( ) \text{ (solve)}$$

FN % multiply it by 100.

10.  $Z \sim N(0, 1)$ ;  $Z = \frac{X - \mu}{\sigma}$ ,  $X \sim N(\mu, \sigma^2)$

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}; -\infty < z < \infty$$

$X$ : Life of army shoes

$$X \sim N(\mu, \sigma^2) \quad \mu = 8 \text{ months}, \sigma = 2 \text{ months}$$

$$N = 5000$$

$$P(\text{A pair of shoes working after 12 months}) = P(X > 12)$$

$$\begin{aligned} P(\text{A pair of shoes need replacement after 12 months}) \\ = 1 - P(\text{A pair of shoes working after 12 months}) \\ = 1 - P(X > 12) \end{aligned}$$

No. of pairs of shoes need replacement after 12 months:

$$f(X) = N[1 - P(X > 12)]$$

$$= 5000 \times [1 - P(X > 12)] = ( ) \text{ solve it.}$$

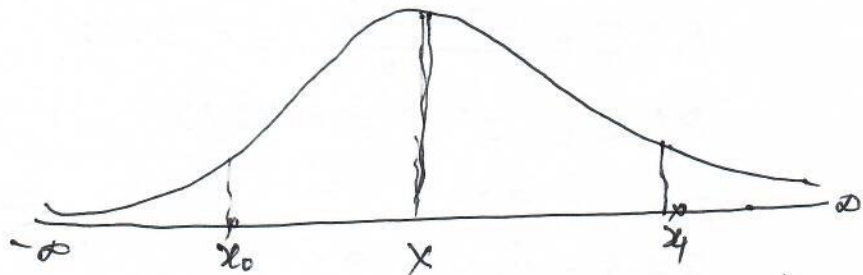
11.

$$N = 200 \quad X: \text{Wages} \quad X \sim N(\mu, \sigma^2)$$

$$\mu = 200, \sigma^2 = 400, \sigma = 20$$

$$P(\text{highest paid 197 workers}) = \frac{197}{200} = 0.985$$

$$P(\text{lowest paid 197 workers}) = \frac{197}{200} = 0.985$$



$x_0$  is the lowest paid wage of 197 highest paid workers  
 $x_1$  is the highest paid wage of 197 lowest paid workers.

$$P(X \leq x_1) = 0.985 \quad \text{and} \quad P(X > x_0) = 0.985$$

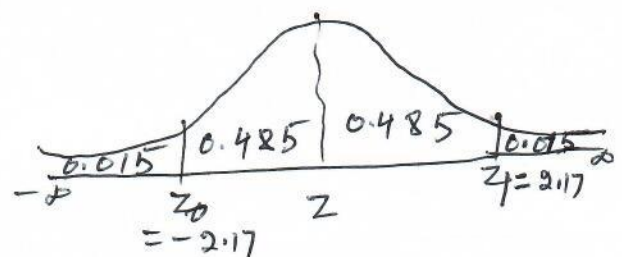
$$P(Z \leq \frac{x_1 - 200}{20}) = 0.985 \quad \text{and} \quad P(Z > \frac{x_0 - 200}{20}) = 0.985$$

$$P(Z \leq z_1) = 0.985 \quad \text{and} \quad P(Z > z_0) = 0.985$$

$$z_1 = \frac{x_1 - 200}{20} \quad \text{and} \quad z_0 = \frac{x_0 - 200}{20}$$

Hence  $\frac{x_1 - 200}{20} = 2.17$  (Gives  $x_1$ )

&  $\frac{x_0 - 200}{20} = -2.17$  (Gives  $x_0$ )





12

$$X \sim N(\mu, \sigma^2); \mu = 800, \sigma = 150$$

$$N = 50,000$$

$$i) f(X \leq 650) = N P(X \leq 650)$$

$$ii) f(X \geq 950) = N P(X \geq 950)$$

Solve using Area under normal curve table.

13

$$X \sim N(\mu, \sigma^2), \text{ Given } \mu = 12 \text{ cm}, \sigma = 0.02 \text{ cm}$$

$$i) P(X \geq 12.05) = (\text{solve})$$

$$ii) P(X \geq K) = 0.90 \text{ (Solve for } K)$$

$$iii) P(11.95 \leq X \leq 12.05) = (\text{solve})$$

14 → 13

15 → 11

16

$$X \sim N(\mu, \sigma^2) \quad \mu = ? \quad \sigma^2 = 100, \sigma = 10$$

$$P(X \leq 82.5) = 0.8212, P(X > 58.3) ?$$

First we get the value of  $\mu$ , Given

$$P(X \leq 82.5) = 0.8212$$

$$P\left(Z \leq \frac{82.5 - \mu}{10}\right) = 0.8212$$

$$P(Z \leq Z^*) = 0.8212$$

$$Z^* = 0.92$$

$$\frac{82.5 - \mu}{10} = 0.92$$

$$\mu = 82.5 - 9.2 = 73.3$$

$$\therefore P(X > 58.3) = P\left(Z > \frac{58.3 - 73.3}{10}\right) = P(Z > -1.5)$$

$$\therefore P(Z > -1.5) = 0.5 + P(0 < Z < 1.5)$$

$$= 0.5 + 0.4332$$

$$= 0.9332$$

Ans.

